

5.2 BLIP and Captioning

CoCa adds a decoder. BLIP also filters the web: ITC, ITM, and a language-model loss.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

BLIP (Li et al., 2022) is a vision–language model that **matches** and **writes**. CLIP scores pairs. CoCa (Yu et al.) is the sibling idea: add a **decoder**. BLIP does that, and it also **cleans** the noisy web pairs CLIP trained on.

First time this method appears. BLIP is CLIP-style scoring **plus** a decoder that writes text, with a filter on noisy web pairs.

What. Three losses: ITC (contrastive), ITM (matched vs unmatched pair), LM (caption tokens).

Bootstrap: generate captions, filter, train. **Why.** CLIP cannot decode a sentence. Naive captioners

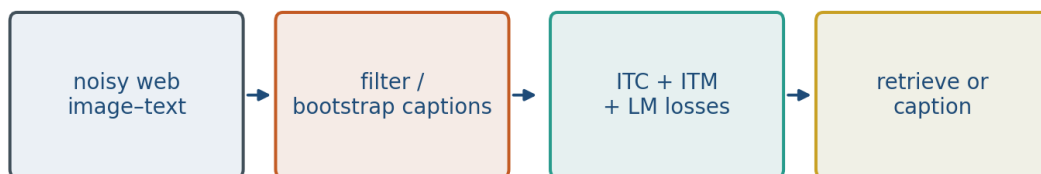
train on noisy alt-text. BLIP tries to clean the web and add generation. **Architecture.** Image encoder, text encoder, and a decoder with **cross-attention** into image tokens (queries from text, keys/values from vision). **How.** ITC aligns. ITM is a binary match head. LM is next-token on the caption.

Generation uses the decoder; retrieval can use ITC embeddings. **Formula.** ITC as InfoNCE; ITM as logistic on a CLS; LM as $\sum_t -\log p(w_t | w_{<t}, \text{image})$. **Tradeoffs.** + Captions and retrieval in one family. – Hallucinated captions; filter is extra machinery; heavier than frozen CLIP.

1. Bootstrap the data

Web alt-text is messy. BLIP’s captioner generates synthetic captions; a filter keeps the ones that still match the image (ITM: image–text matching). The cleaned pairs retrain the model. That loop is the “bootstrapped pretraining” in the syllabus.

BLIP: clean the pairs, then learn to match and to write



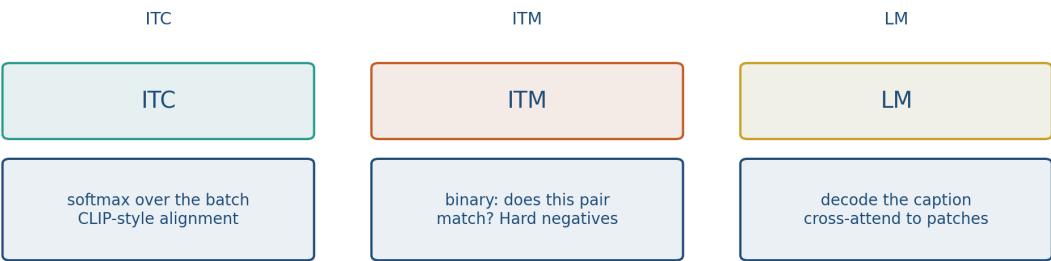
Alt-text like `IMG_4032` or a product SKU is a bad caption. Generating a sentence and then asking “does this still match?” is the filter. You will not run that loop in lab; you will remember it when your project scrapes the web.

CoCa (Yu et al.) adds a captioning loss on top of contrastive training. It does **not** advertise the bootstrap filter. If your story is “noisy web text,” you need BLIP’s ITM filter, not only a decoder.

2. Three losses, one backbone

Loss	Job
ITC (contrastive)	CLIP-style alignment
ITM (matching)	Binary: does this caption belong to this image?
LM (language modeling)	Decode the caption, causal on text, cross-attend to the image

BLIP keeps three losses. CoCa (CS231N) adds a decoder; BLIP also filters the web.



The image tower is a ViT. The text side is a transformer that can encode or decode depending on the head. **BLIP-2** later freezes a strong image encoder and trains a thin **Q-Former**; treat that as “same idea, cheaper.” LLaVA is a frozen CLIP tower plus an LLM. Not this hour.

ITC needs a batch of negatives. ITM can use a **hard** negative (a caption that almost matches). LM is why BLIP can caption at all.

Whisper, which you will watch as an analog, is an encoder–decoder for **speech** → **text**. BLIP is encoder–decoder for **image** → **text**, plus ITC and ITM.

3. Captioning versus retrieval

Retrieval (CLIP) returns an existing string from a gallery. Captioning (BLIP) **generates** a new string. Evaluation: CIDEr / CLIP-score / human, not just recall@k. Hallucinated objects are the usual failure (“a red bike” when there is none).

CLIP-score as a caption metric is circular if your captioner was trained to match CLIP. Prefer at least one human or task-specific check. CIDEr rewards n-gram overlap with reference captions; it punishes valid paraphrases. Report both and show two example captions, one good and one hallucinated.

4. Teaching this note

30–40 minutes. CLIP retrieves vs BLIP writes, then the three-loss table, then one numeric ITM vs ITC contrast. Reading: the BLIP paper (ITC+ITM+LM). Play the Whisper video as the **encoder–decoder analog** for captioning (**0:00–8:00**). Say explicitly: Whisper does not add ITC+ITM; **BLIP does**.

Minute plan: 8 min retrieve vs generate; 10 min ITC/ITM/LM table; 8 min ITM numbers; 12 min Whisper clip plus the one-sentence BLIP add-on.

5. Worked example

One image, two captions: true “a red mug on a desk,” false “a blue bicycle.”

ITC ($N = 2$): softmax over the batch; the false caption is one negative among many in a real batch.

ITM: a classifier on the fused `[CLS]` with labels 1 and 0. Suppose logits $s_+ = 2$, $s_- = -1$.

$$\sigma(s) = \frac{1}{1 + e^{-s}}, \quad \sigma(2) \approx 0.88, \quad \sigma(-1) \approx 0.27.$$

Binary cross-entropy: matched term $-\log \sigma(2) \approx 0.13$; mismatched term $-\log(1 - \sigma(-1)) \approx 0.31$.

ITM is not a batch softmax. One pair, a yes/no score.

ITM logit	sigmoid	binary CE term
matched $s = 2$	$\sigma(2) \approx 0.88$	$-\log \sigma \approx 0.13$
mismatch $s = -1$	$\sigma(-1) \approx 0.27$	$-\log(1 - \sigma) \approx 0.31$

LM: decode `a , red , mug , ...` with cross-attention into ViT patch tokens. Loss is next-token, like GPT, but the “prefix” includes the image.

If alt-text was `DSC0001.jpg`, LM on that string teaches garbage. Filter first, then LM.

ITC wants a **batch**. ITM can score one pair. LM wants a **token sequence**. If your project only retrieves, stop at ITC (or CLIP). If it must write, you need LM. If web text is junk, you need the filter (ITM).

6. Where students get stuck

- Using recall@k as the only captioning metric.
- Thinking ITM and ITC are the same softmax.

- Skipping the filter and training LM on raw alt-text.
 - Calling CoCa and BLIP the same paper. Shared idea: a decoder. BLIP's extra: bootstrap + ITM.
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7. Video

Keep the **BLIP paper** as the reading (Li et al., 2022). For a lecture clip, watch [OpenAI's Whisper Model Explained](#) as an **encoder-decoder analog for captioning**: audio encoder, text decoder. Pause on that cross-attention picture.

Then say: **BLIP adds ITC + ITM + LM** on image-text pairs, and bootstraps noisy web captions. Whisper transcribes; BLIP captions and matches.

8. Practice

1. Why filter generated captions instead of trusting every alt-text on the web?
2. ITM vs ITC: one is a binary matcher, one is a softmax over the batch. When would ITM catch a pair ITC misses?
3. Name one metric you would report in a captioning project that CLIP retrieval would not need.
4. ITM logits $s = 0$ vs $s = 2$ for a matched pair. Compute $\sigma(s)$ in both cases. Write the binary-CE term $-\log\sigma(s)$.
5. A caption of 6 tokens (including `EOS`). LM loss is mean token NLL. If five tokens have probability 0.5 and `EOS` has probability 0.25, what is the mean NLL? (Use log base e : $-\log 0.5 \approx 0.69$, $-\log 0.25 \approx 1.39$.)